Client Feature Requirements — Database & Table Impact

Product: Ecommerce Guru (Listing Analyzer Pro)
Document purpose: Map each requested client feature to **existing database tables** (additive changes only) and **new tables**, without breaking current behavior.
Status: Planning only — **not implemented** until client/stakeholder approval.
Prepared: 26 October 2026 (IST)

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1. What this document is for

Your client asked for seven capability areas (Excel attributes, categories, bulk content, multi-marketplace content, revenue calculator, A+ graphics, policy-compliant AI). This document answers:

- What already exists in the app today
- Which **PostgreSQL tables** would change (additive columns only where noted)
- Which **new tables** would be added
- How each change ties to a **feature**, in plain language

Principle: Existing flows keep working. New fields are optional. Current `audits.generated_content`, export columns, and single-product "Generate content" stay the default path.

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2. Current database touchpoints (baseline)

These tables already support listing work today:

Table	Role today
`audits`	Product project: name, category (text), title, bullets, keywords, images, `generated_content`, `content_marketplace_id`, `generated_images` (includes A+), scores, workflow step
`content_marketplaces`	Per-marketplace AI instructions, rules (JSON), enabled flag
`product_profiles`	SKU, priority, manager, `target_marketplaces` (JSON array), workflow template
`graphics_projects`	Separate "Create Graphics" projects (lifestyle/feature images)
`image_prompt_templates`	Admin prompts for **graphics** and **aplus** image generation
`product_orders`	Synced orders → revenue display on product detail (not fee calculator)
`settings`	Admin key/value config (email, platform, Amazon, etc.)
`amazon_seller_connections`	Amazon OAuth / SP-API connection
`amazon_publish_jobs`	Publish job history per audit

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3. Feature map → table changes

Feature A — Additional listing attributes (Excel template)

Client ask: More attribute columns in Excel for complete product data (per reference file).

Today: Export uses ~25 Amazon flat-file-style columns; `feed_product_type` is empty; category goes to `item_type` only.

Impact type	Table / object	Change
Modify (additive)	`audits`	Add optional `listing_attributes` (jsonb) — e.g. material, color, size, compliance fields. Add optional `feed_product_type` (text) or store inside json.
Modify (additive)	`product_profiles`	Optional defaults: e.g. preferred SKU pattern, default attribute values (jsonb) if needed.
New	`amazon_listing_attribute_templates`	Defines columns per product type: field key, label, max length, required, Excel column name.
New	`amazon_product_types`	Catalog: marketplace (IN/US/...), feed product type code, display name, link to attribute template.

No breaking change: Existing Excel columns remain; new columns appended or second sheet “Attributes”. Empty json = same export as today.

App layers (for reference, not DB): `amazon-listing-export.ts` , Products bulk export API, audit upload step UI.

Feature B — Category expansion (incl. Amazon India / fees context)

Client ask: More product categories aligned with listing requirements (ref: Amazon India fees & pricing).

Today: Long hardcoded category list in UI; stored as free text on `audits.category` .

Impact type	Table / object	Change
Modify (additive)	`audits`	Optional `amazon_product_type_id` (FK) or `category_metadata` (jsonb): browse node, marketplace, fee category hint. Keep existing `category` text for backward compatibility.
New	`amazon_product_types`	Same as Feature A — shared catalog for categories / feed types / marketplace.
Optional new	`amazon_marketplace_fee_hints`	Lightweight link: product type + marketplace → default referral % (feeds Feature E).

No breaking change: Old projects keep text `category` ; new picker can dual-write text + structured id.

App layers: Audit workflow category dropdown, export `item_type` / `feed_product_type` .

Feature C — Bulk content creation

Client ask: Generate listing content for many products at once.

Today: Bulk **Excel export** only (up to 100 products). AI generation is **one audit at a time**.

Impact type	Table / object	Change
New	`content_generation_jobs`	Job header: `user_id`, `workspace_id`, status, total/succeeded/failed counts, created/completed timestamps, optional error summary.
New	`content_generation_job_items`	Per audit: `job_id`, `audit_id`, status, error message, credits charged, timestamps.
Unchanged (writes)	`audits`	Still updates `generated_content` (and optionally variants — see Feature D) per item, same as today's single generate.

No breaking change: Existing `/audits/:id/generate-content` unchanged; bulk is a new API + Products UI action.

App layers: New routes, credit preflight sum, notifications optional (`notifications` table may receive rows — already exists).

Feature D — Multi-marketplace content generation (simultaneous)

Client ask: Generate content for multiple marketplaces at once; Excel per reference sheet.

Today: One `content_marketplace_id` and one `generated_content` blob per audit.

Impact type	Table / object	Change
Modify (none required)	`audits`	Keep `generated_content` + `content_marketplace_id` as **primary** copy (default marketplace).
New	`audit_content_variants`	Columns: `audit_id`, `content_marketplace_id` (or slug), `generated_content` (jsonb), `updated_at`. Unique (`audit_id`, marketplace).
Modify (optional seed)	`content_marketplaces`	Add row e.g. `amazon-in` if India rules differ from generic `amazon` slug.

No breaking change: Preview/export default to primary `audits.generated_content` until user picks a variant.

App layers: Multi-select on content step, batch call to `generateListingContent` , bulk Excel multi-sheet export.

Feature E — Revenue calculator

Client ask: Estimate revenue, fees, and profitability (Amazon India fees page as reference).

Today: `product_orders` shows **historical** revenue from connected stores — not referral/FBA/closing fee math.

Impact type	Table / object	Change
New	`marketplace_fee_rules`	Marketplace code, category/product type, referral %, min/max fee, closing fee, currency, optional FBA size tier rules (jsonb), effective_from date.
New (optional)	`product_profit_scenarios`	Save calculator inputs/outputs per `audit_id` or user: price, COGS, dimensions, computed margin (jsonb).
Unchanged	`product_orders`	Historical orders unchanged.
Unchanged	`audits`	No requirement to change unless saving scenario on audit.

No breaking change: Pure new page/tool; optional "save scenario" only.

App layers: New calculator UI; admin screen to refresh fee tables (or seed from CSV).

Feature F — A+ Content graphics

Client ask: Create/generate graphics for Amazon A+ Content (reference doc).

Today: A+ modules, image generation, storage in `audits.generated_images` → `aplus`; templates in `image_prompt_templates` (`category = aplus`).

Impact type	Table / object	Change
Modify (data only)	`image_prompt_templates`	New/updated template rows from client doc (no schema break).
Modify (optional additive)	`image_prompt_templates`	Optional new jsonb metadata keys in existing `metadata` column (module layout, dimensions).
Modify (json only)	`audits`	`generated_images.aplus` may gain new module ids; old audits keep old shape.
Unchanged	`graphics_projects`	Main listing graphics remain separate unless you explicitly link A+ to a graphics project later.

No breaking change: Existing A+ projects continue to load; new modules are additive.

App layers: `aplus-generator`, audit workflow A+ step, admin Image Prompt Templates.

Feature G — Amazon listing policy guidelines (AI content)

Client ask: AI output must follow marketplace policy documents (reference docs).

Today: Long static Amazon instructions in code + `content_marketplaces.ai_instructions`; admin can edit Marketplace Content settings.

Impact type	Table / object	Change

Modify (additive)	`content_marketplaces`	Optional `policy_version`, `policy_excerpt` (text), or `policy_document_id` (FK).
New	`marketplace_policy_documents`	`id`, marketplace slug/category, title, body text or storage URL, version, `is_active`, timestamps.
Modify (optional)	`settings`	Global "active policy pack" version key under category e.g. `amazon_compliance`.
Unchanged	`audits`	Optional future: `result` json could include policy compliance score — not required for v1.

No breaking change: If no policy doc attached, prompts behave as today.

App layers: `content-generator.ts`, `content-marketplace-service.ts`, admin upload/paste UI.

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4. Consolidated table checklist

Existing tables — additive changes only

Table	Features	Proposed new columns / notes
`audits`	A, B, D, (F json)	`listing_attributes` jsonb; `feed_product_type` text; `amazon_product_type_id` or `category_metadata` jsonb; `generated_images` extended in app only unless new column needed
`product_profiles`	A	Optional attribute/SKU defaults jsonb
`content_marketplaces`	D, G	Optional India row; policy fields or FK to policy docs
`image_prompt_templates`	F	More seed rows; optional metadata keys
`settings`	G	Optional compliance version keys

New tables

Table	Features	Purpose
`amazon_product_types`	A, B	Product type / category catalog per marketplace
`amazon_listing_attribute_templates`	A	Column definitions per product type for Excel + UI
`audit_content_variants`	D	Per-marketplace generated copy
`content_generation_jobs`	C	Bulk generate job header
`content_generation_job_items`	C	Per-audit bulk job line items
`marketplace_fee_rules`	E	Referral/closing/FBA fee data
`product_profit_scenarios`	E (optional)	Saved calculator runs

`marketplace_policy_documents`	G	Versioned policy text for AI prompts
`amazon_marketplace_fee_hints`	B, E (optional)	Link types to default fee %

Tables explicitly unchanged (for these features)

`auth_*`, `credits`, `plans`, `workspaces`, `team_members`, `graphics_projects` (unless later A+ link), `amazon_seller_connections`, `amazon_publish_jobs`, `product_orders`, `blog_posts`, `sellermate_*`, etc.

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5. Suggested implementation phases (for approval)

Phase	Features	DB scope
1	A + B + G (partial)	`amazon_product_types`, `amazon_listing_attribute_templates`, `audits` attributes, policy docs
2	D	`audit_content_variants`, export multi-sheet
3	C	Job tables + bulk UI
4	E	Fee rules + calculator UI
5	F	Template/content from client A+ doc (mostly data)

Each phase can ship independently without removing prior behavior.

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6. Approvals required before development

- [ ] Confirm feature priority and phase order
- [ ] Provide reference Excel + policy + A+ documents (exact columns and modules)
- [ ] Confirm multi-marketplace list (e.g. Amazon IN + US + eBay only)
- [ ] Confirm revenue calculator scope (India-first vs all export marketplaces)
- [ ] Approve this table plan or request edits

No database migrations or application code will be applied until written approval is given.

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7. Document control

Version	Date	Notes
1.0	26 Oct 2026	Initial planning document from codebase analysis

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End of document